CLASS 8 · ICSE · SELINA CONCISE MATHEMATICS · SHORT NOTES

Chapter 19: Representing 3-D in 2-D

Topic-wise short notes with solved examples

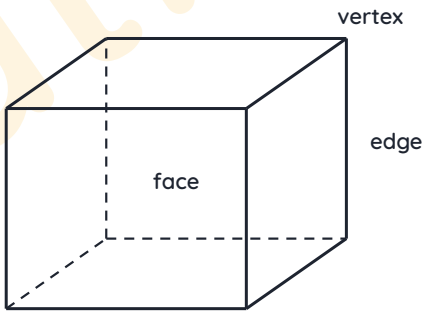
Solid shapes and their faces, edges and vertices, Euler's formula, nets of solids, and the different views of a three-dimensional object.

1 Two and three dimensions

- A **plane (2-D) figure** has length and breadth only: triangle, rectangle, circle.
- A **solid (3-D) figure** has length, breadth and height: cube, cylinder, sphere.
- On paper a solid can be shown by an **oblique sketch** (on squared paper, not to exact scale) or an **isometric sketch** (on isometric dot paper, with true lengths along three directions).

2 Faces, edges and vertices

- A **polyhedron** is a solid bounded only by flat polygonal faces.
- **Face**: a flat surface. **Edge**: a line segment where two faces meet. **Vertex**: a point where three or more edges meet.
- Cylinders, cones and spheres have curved surfaces, so they are **not** polyhedra.



A cuboid has 6 faces, 12 edges and 8 vertices

Solid	Faces (F)	Vertices (V)	Edges (E)
Cube	6	8	12
Cuboid	6	8	12
Triangular prism	5	6	9
Triangular pyramid (tetrahedron)	4	4	6
Square pyramid	5	5	8
Pentagonal prism	7	10	15
Hexagonal pyramid	7	7	12

3 Prisms and pyramids

- A **prism** has two congruent, parallel polygonal ends joined by rectangular (or parallelogram) faces. It is named after its ends.
- A **pyramid** has a polygonal base and triangular faces meeting at one vertex (the apex). It is named after its base.

Solid with an n-sided base	Faces	Vertices	Edges
Prism	$n+2$	$2n$	$3n$
Pyramid	$n+1$	$n+1$	$2n$

Example 3.1 Find the number of faces, vertices and edges of an octagonal prism.

SOLUTION

$n = 8$

10 faces, 16 vertices, 24 edges

Example 3.2 Find the number of faces, vertices and edges of a pentagonal pyramid.

SOLUTION

6 faces, 6 vertices, 10 edges

4 Euler's formula

- For every polyhedron:

$$F + V - E = 2$$

- It can be used to find one of the three numbers when the other two are known, or to check whether a polyhedron with given numbers can exist.

Example 4.1 A polyhedron has 8 faces and 12 vertices. How many edges does it have?

SOLUTION

$8 + 12 - E = 2$

18

Example 4.2 A polyhedron has 20 edges and 12 vertices. Find the number of faces.

SOLUTION

$F + 12 - 20 = 2$

10

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Example 4.3 Can a polyhedron have 10 faces, 20 edges and 15 vertices?

SOLUTION

$10 + 15 - 20 = 5$, which is not 2.

No

Example 4.4 Verify Euler's formula for a square pyramid.

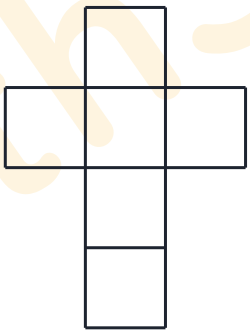
SOLUTION

$F = 5, V = 5, E = 8$: $5 + 5 - 8$

2, so the formula holds

5 Nets

- A **net** is a plane figure that can be folded to form a solid.
- A cube has 6 square faces, so each of its nets has 6 squares. There are 11 different nets of a cube.
- Not every arrangement of 6 squares is a net: four squares meeting at a point, or six squares in one row, will not fold into a cube.



A net of a cube

Solid	Its net is made of
Cube	6 squares
Cuboid	6 rectangles in three equal pairs
Triangular prism	2 triangles and 3 rectangles
Square pyramid	1 square and 4 triangles
Tetrahedron	4 triangles
Cylinder	2 circles and 1 rectangle
Cone	1 circle and 1 sector of a circle

- In a **die**, the numbers on opposite faces add up to 7: 1 and 6, 2 and 5, 3 and 4. In a net of a die, opposite faces are never next to each other.

Example 5.1 In the cross-shaped net of a cube shown above, the four squares in the column are numbered 1, 2, 3, 4 from the top and the side squares are 5 (left) and 6 (right). Which faces are opposite each other?

SOLUTION

In a column of four, the first and third are opposite, and so are the second and fourth. The two side squares are opposite each other.

1 and 3; 2 and 4; 5 and 6

Example 5.2 What solid is formed by a net of two equal circles and a rectangle?

SOLUTION

A cylinder

6 Views of a solid

- A solid looks different from different directions. The three standard views are the **front view**, the **side view** and the **top view** (plan).

Solid (standing upright)	Front view	Side view	Top view
Cube	Square	Square	Square
Cuboid	Rectangle	Rectangle	Rectangle
Cylinder	Rectangle	Rectangle	Circle
Cone	Triangle	Triangle	Circle with its centre marked
Sphere	Circle	Circle	Circle
Square pyramid	Triangle	Triangle	Square with its diagonals

- A **cross-section** is the shape obtained when a solid is cut by a plane. A cube cut parallel to a face gives a square; a cylinder cut parallel to its base gives a circle, and cut vertically through its axis gives a rectangle.

Example 6.1 Three cubes are placed in a row. Describe the front, side and top views.

SOLUTION

Front: 3 squares in a row. Side: 1 square. Top: 3 squares in a row.

Example 6.2 What is the cross-section when a cone is cut by a plane parallel to its base?

SOLUTION

A circle

7 Common mistakes

- Counting hidden edges or vertices twice, or missing them.
- Applying Euler's formula to a cylinder, cone or sphere.
- Mixing up the formulas for prisms and pyramids.
- Assuming that any six joined squares form a net of a cube.
- Confusing the top view with the front view.
- Writing Euler's formula as $F + E - V = 2$.

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